



Ammo visualisation

Google made this wonderful visualisation of arms and ammo import and export across countries. <http://dgit.in/Ripe9m>

Big Data Goes official

The Obama Administration announced the Big Data Research and Development Initiative, to understand use of big data in troubleshooting govt problems

Big data = Big opportunities?

All those bits and bytes
do add up to a lot

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The 2002 science-fiction movie *Minority Report* showed a special police unit that is able to arrest murderers before they commit their crimes. In another case of science-fiction turning into science-fact, a team led by well-known technology innovators Donnie Fowler and Zach Friend in association with the Santa Cruz (California) police department, have managed to create a similar set-up. However, rather than relying on psychic humans, they decided to harness the power of something called Big Data. This new field of study enabled them to mine vast amounts of crime data from the city, and use it to set up a system that, quite frankly, can only be described as uber-cool. Called “predictive policing” or PredPol, the system has seen great success in the iconic city of Santa Cruz, with property theft dropping 19 per cent in the first six months of 2012, compared to the same timeframe in 2011, when the technology was not being used. This, without adding more officers to the workforce, or changing beat and shift timings.

What is Big Data?

Wikipedia has the following to say about Big Data, “In information technology, big data is a collection of data sets so large and complex that it becomes difficult to process using on-hand database management tools. The challenges include capture, curation, storage, search, sharing, analysis, and visualization.” Straightforward enough. Essentially, Big Data means any collection/segment of data so large that analyzing it using “regular” data management systems becomes nigh impossible.

But how large is “too large”? For most people, a few terabytes is more than they can handle. For a small or medium-sized firm, 300-500 TB might be the limit that



Meet Big Data, the next big thing in Information Technology

stretches their data management capacity. What about very large global outfits and multinational corporations? By recent estimates, Google processes about 24 petabytes of data in a day (1 petabyte = 1000 terabytes = 1 million gigabytes). That means by the end of 2012, Google would have processed roughly 8760 petabytes of data. Such astronomical figures are beyond the comprehension of most people, yet these are essentially what form the basic premise of Big Data.

DATA = CURRENCY?

- “Data is the new oil.” Coined in 2006 by Clive Humby, a British data commercialization entrepreneur, this now famous phrase was embraced by the World Economic Forum in a 2011 report, which considered data to be an important economic asset, like oil.
- From the beginning of recorded history until 2003, we created 5 exabytes (5 billion gigabytes) of data. In 2011, the same amount was created every two days. By 2013, it's expected that the time will shrink to 10 minutes.

Big Data definitions are therefore quite subjective, depending on the communities/organizations concerned. However, all definitions are based on three basic parameters: Volume (how much data...), Variety (...of what kinds...) and Velocity (...is received at what speed.) In recent times, data analysts have also agreed to include Virtual as a parameter, thereby largely restricting the definition of data/information to what is stored digitally in filesystems and data systems globally.

Why should you care?

Most of the times, Big Data is useful only to large corporations and companies looking to better understand consumer behaviour and market trends, and utilize that knowledge to smoothen their work practises, cater to the precise needs and tastes of the markets, and make a tidy profit along the way. After all, an average Joe has no use for the kind of knowledge Big Data can furnish, correct? Well, both right, and wrong.

Big Data in fact can be very useful to even the common man not interested in extensive number crunching, by simply